DHCP Setup

BY

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Introduction

Dynamic Host Configuration Protocol (DHCP) is an essential network protocol used to automatically provide devices with the network configuration required for communication. Instead of manually assigning an IP address, subnet mask, default gateway, and DNS server to every device, a DHCP server assigns these settings automatically. In this practical setup, a router, switch, server, and end devices are connected to create a network where the server acts as the DHCP server and automatically assigns IP addresses to the connected PCs.

DHCP

DHCP stands for Dynamic Host Configuration Protocol, a network management protocol that automatically assigns IP addresses and configuration settings to devices on a network.

What DHCP Does

• Assigns IP Addresses: Gives each connected device a unique temporary IP address.

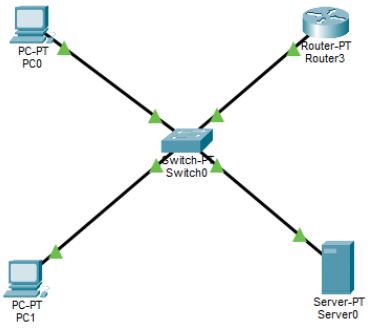
• Subnet Mask: Defines the range of the local network.

• Default Gateway: Provides the router address to reach outside networks and the internet.

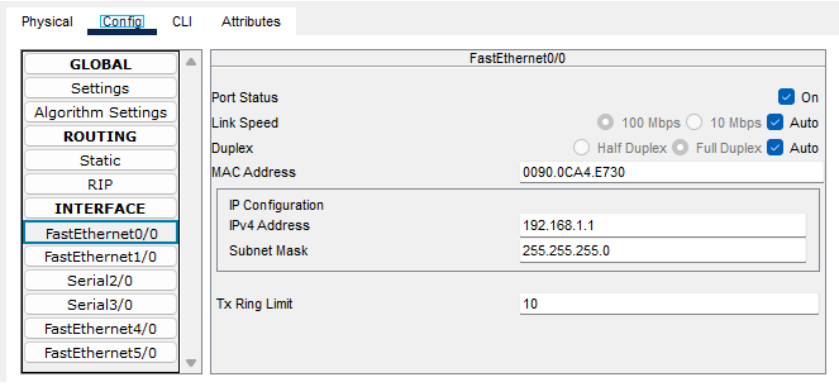
• DNS Servers: Supplies domain name server addresses to translate website names into IP addresses.

DHCP Setup

To setup DHCP, we need end devices, router (Default Gateway), switch, and a server to configure IP automatically. Connect them according to the SS below:

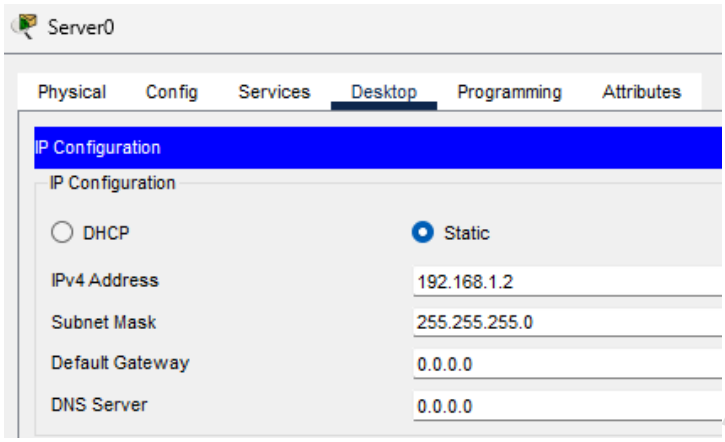


Then we need to assign the router an IP which will function as a default gateway for the communication in the network and outside the network.



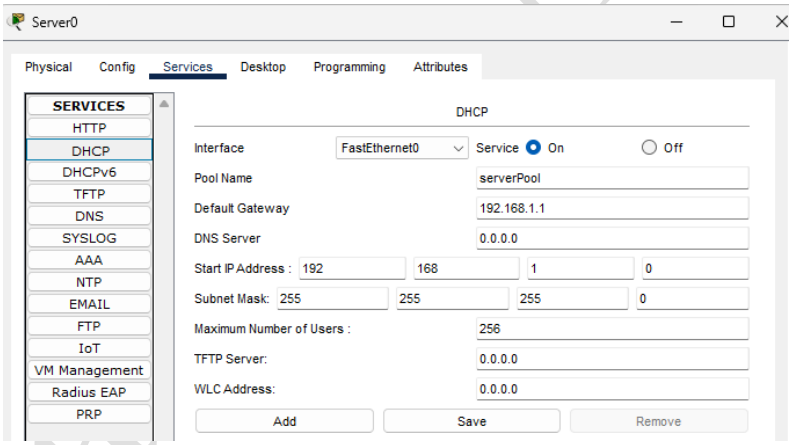
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After assigning the IP to the router we assign IP to the server as we need the server to have an IP before to give other devices in the network an IP. After that we need to enable the DHCP configuration.

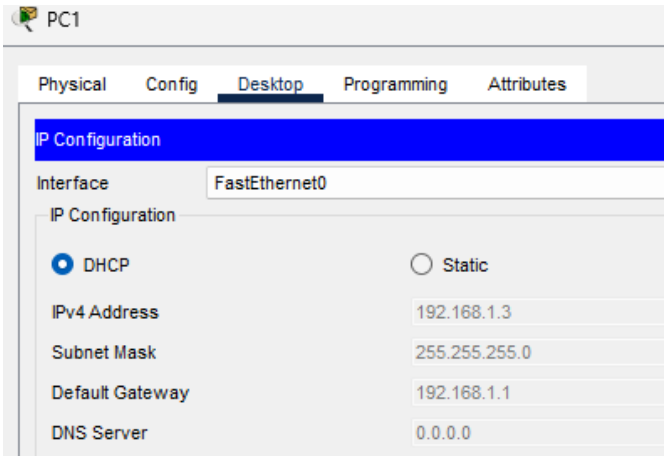


DHCP Configuration

To enable the DHCP configuration, we need to open the server and in services we can see various options, select DHCP and enter the default gateway, whatever IP is assigned to the router. Then enter the series of IP you want the network to have in the **Start IP Address**. The Server then will check what IP's are already given, i.e. to the router and server.



Then we need to open the PC and enable DHCP in there for the server to assign IP automatically to the end device, i.e. PC.



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Conclusion

This setup demonstrates how DHCP simplifies network configuration by automatically assigning IP addresses and other essential network settings to end devices. By configuring the router as the default gateway and the server as the DHCP server, PCs can obtain their network configuration automatically without manual IP assignment. This makes DHCP useful for managing networks efficiently, especially when a large number of devices need to be connected.

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